

Project: Hunter Executive Summary & System Documentation

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Subject: project: Hunter Yonatan Avitan

Executive Summary

The "Hunter" project is an advanced, automated network forensics and threat detection tool developed for the Kali Linux environment. Its primary objective is to monitor live network traffic, identify connections to malicious indicators (IOCs), and analyze suspicious files transferred over the network.

Operating passively, the system dynamically pulls Threat Intelligence Feeds and integrates advanced tools such as Tshark for packet analysis and the VirusTotal API for file signature verification. This tool provides a comprehensive solution for Incident Response (SOC/IR) teams through real-time alerts and accurate, forensically sound event logging.

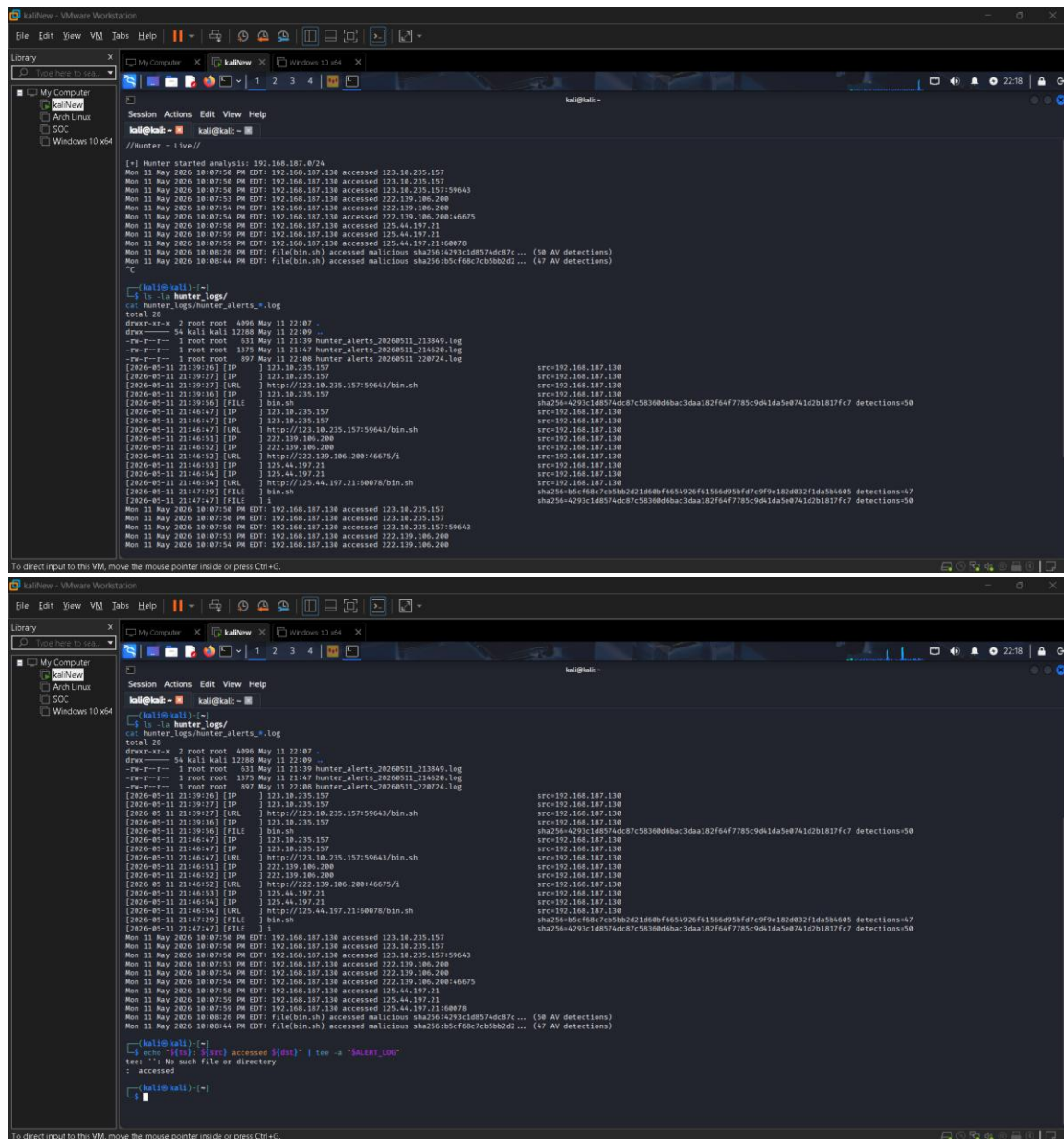
System Functions Documentation

1. Network Analysis This module is responsible for monitoring live network traffic. It utilizes Tshark to extract real-time data such as IP addresses (source and destination), DNS requests, and HTTP hosts. Simultaneously, the function dynamically downloads updated blacklists (Indicators of Compromise) from verified global sources and compares the live traffic against them. When a connection to a malicious IP or domain is detected, the system triggers an immediate alert.

2. File Analysis This function operates in the background, capturing traffic into rotating PCAP files. A dedicated background process reads these files, extracts transferred network objects (such as HTTP files), and filters files smaller than 1MB (per the project requirements). Every file that passes the filter undergoes SHA-256 hashing, and its signature is automatically sent to the VirusTotal API for scanning. Files identified as malicious are saved in an evidence directory for further investigation.

3. Log and Audit A critical component for investigation and forensics. Every anomalous detection (whether a malicious network connection or an infected file) is forwarded to the logging function. This function creates a standardized record containing a precise timestamp, source address, destination address (or file details), and the nature of the threat. The data is securely saved to a central log file, ensuring all events are documented and retained even after the software is closed.

4. Monitoring (Live Monitoring) The Command Line Interface (CLI) of the tool was specifically designed for cybersecurity personnel operating in the field. Upon execution, a clean screen displays the project details and the listening status ("//Hunter Live//"). The system provides continuous, color-coded visual alerts for detected vulnerabilities. Additionally, a "Heartbeat" mechanism is integrated, confirming to the operator that the system is running and scanning properly in the background without cluttering the main alert screen.



```
[*] Hunter started analysis: 192.168.187.0/24
Mon 11 May 2026 10:07:50 PM EDT: 192.168.187.130 accessed 123.10.235.157
Mon 11 May 2026 10:07:50 PM EDT: 192.168.187.130 accessed 123.10.235.157
Mon 11 May 2026 10:07:50 PM EDT: 192.168.187.130 accessed 123.10.235.157/59643
Mon 11 May 2026 10:07:53 PM EDT: 192.168.187.130 accessed 222.139.186.200
Mon 11 May 2026 10:07:54 PM EDT: 192.168.187.130 accessed 222.139.186.200
Mon 11 May 2026 10:07:54 PM EDT: 192.168.187.130 accessed 222.139.186.200/46675
Mon 11 May 2026 10:07:54 PM EDT: 192.168.187.130 accessed 125.44.197.21
Mon 11 May 2026 10:07:59 PM EDT: 192.168.187.130 accessed 125.44.197.21/60078
Mon 11 May 2026 10:08:20 PM EDT: File(bin.sh) accessed malicious sha256:4299c1d8574dc87c... (50 AV detections)
Mon 11 May 2026 10:08:44 PM EDT: File(bin.sh) accessed malicious sha256:b5cf68c7cb5b0b2d... (47 AV detections)

kali@kali:~$ ls -l hunter_logs/
total 28
drwxr-xr-x 2 root root 4096 May 11 22:07 .
drwxr-xr-x 54 kali kali 12288 May 11 22:09 ..
-rw-r--r-- 1 root root 631 May 11 21:39 Hunter_alerts_20260511_213849.log
-rw-r--r-- 1 root root 1375 May 11 21:47 Hunter_alerts_20260511_214620.log
-rw-r--r-- 1 root root 897 May 11 22:08 Hunter_alerts_20260511_220724.log

2026-05-11 21:39:26 [IP] 123.10.235.157
2026-05-11 21:39:27 [IP] 123.10.235.157
2026-05-11 21:39:27 [URL] http://123.10.235.157/59643/bin.sh
2026-05-11 21:39:36 [IP] 123.10.235.157
2026-05-11 21:39:36 [FILE] bin.sh
2026-05-11 21:40:47 [IP] 123.10.235.157
2026-05-11 21:40:47 [IP] 123.10.235.157
2026-05-11 21:40:47 [URL] http://123.10.235.157/59643/bin.sh
2026-05-11 21:40:51 [IP] 222.139.186.200
2026-05-11 21:40:52 [IP] 222.139.186.200
2026-05-11 21:40:52 [URL] http://222.139.186.200/46675/1
2026-05-11 21:40:53 [IP] 125.44.197.21
2026-05-11 21:40:54 [IP] 125.44.197.21
2026-05-11 21:40:54 [URL] http://125.44.197.21/60078/bin.sh
2026-05-11 21:47:29 [FILE] bin.sh
2026-05-11 21:47:47 [FILE] 1
Mon 11 May 2026 10:07:50 PM EDT: 192.168.187.130 accessed 123.10.235.157
Mon 11 May 2026 10:07:50 PM EDT: 192.168.187.130 accessed 123.10.235.157
Mon 11 May 2026 10:07:50 PM EDT: 192.168.187.130 accessed 123.10.235.157/59643
Mon 11 May 2026 10:07:53 PM EDT: 192.168.187.130 accessed 222.139.186.200
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kali@kali:~$ echo $(cat /dev/urandom | tr -dc 'A' | fold -w 50 | tee -a "/dev/null
tee: -: No such file or directory
: accessed

kali@kali:~$
```